In [234…
```
import pandas as pd
import numpy as np
```

In [235…
```
data = pd.read_csv("/Users/lenovo/Desktop/A2.csv")
```

In [236…
```
df = pd.DataFrame(data)
df
```

Out[236…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [237…
```
df.isnull()
```

Out[237…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	False	False	False	False	False	False	False
1	False	False	False	False	False	True	False
2	False	False	False	False	False	False	False
3	False	False	False	False	True	False	False
4	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...
996	False	False	False	False	False	False	False
997	False	False	False	False	False	False	False
998	False	False	False	False	False	False	False
999	False	False	False	False	False	False	False
1000	False	False	False	False	False	False	False

1001 rows × 7 columns

In [238…

```
df.notnull()
```

Out[238…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	True	True	True	True	True	True	True
1	True	True	True	True	True	False	True
2	True	True	True	True	True	True	True
3	True	True	True	True	False	True	True
4	True	True	True	True	True	True	True
...	...	...	...	...	...	...	...
996	True	True	True	True	True	True	True
997	True	True	True	True	True	True	True
998	True	True	True	True	True	True	True
999	True	True	True	True	True	True	True
1000	True	True	True	True	True	True	True

1001 rows × 7 columns

In [239…

```
df.isnull().sum()
```

Out[239…

```
Roll No       0
Name          0
Subject 1     0
Subject 2     0
Subject 3     1
Subject 4     1
Attendance    0
dtype: int64
```

In [240…

```
df.notnull().sum()
```

Out[240…
```
Roll No       1001
Name          1001
Subject 1     1001
Subject 2     1001
Subject 3     1000
Subject 4     1000
Attendance    1001
dtype: int64
```

In [241…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1001 entries, 0 to 1000
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Roll No     1001 non-null   int64
 1   Name        1001 non-null   object
 2   Subject 1   1001 non-null   int64
 3   Subject 2   1001 non-null   int64
 4   Subject 3   1000 non-null   float64
 5   Subject 4   1000 non-null   float64
 6   Attendance  1001 non-null   int64
dtypes: float64(2), int64(4), object(1)
memory usage: 54.9+ KB
```

In [242…
```
pd.notnull(df)
```

Out[242…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	True	True	True	True	True	True	True
1	True	True	True	True	True	False	True
2	True	True	True	True	True	True	True
3	True	True	True	True	False	True	True
4	True	True	True	True	True	True	True
...	...	...	...	...	...	...	...
996	True	True	True	True	True	True	True
997	True	True	True	True	True	True	True
998	True	True	True	True	True	True	True
999	True	True	True	True	True	True	True
1000	True	True	True	True	True	True	True

1001 rows × 7 columns

In [243…

```
df.fillna(0,inplace=True)
df
```

Out[243…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	0.0	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	0.0	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [244…

```
data1 = pd.read_csv("/Users/lenovo/Desktop/A2.csv")
```

In [245…

```
df1 = pd.DataFrame(data1)
df1
```

Out[245…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [246…

```
df1.isnull().sum()
```

Out[246…

```
Roll No       0
Name          0
Subject 1     0
Subject 2     0
Subject 3     1
Subject 4     1
Attendance    0
dtype: int64
```

In [247…

```
df1.fillna(10,inplace=True)
df1
```

Out[247…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	10.0	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	10.0	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [248…

```
df1.isnull().sum()
```

Out[248…

```
Roll No       0
Name          0
Subject 1     0
Subject 2     0
Subject 3     0
Subject 4     0
Attendance    0
dtype: int64
```

In [249…

```
data = pd.read_csv("/Users/lenovo/Desktop/A2.csv")
```

In [250…

```
df2 = pd.DataFrame(data)
df2
```

Out[250…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [251…

```
df2.ffill()
```

Out[251…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	92.0	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	71.0	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [252…

```
df3 = pd.DataFrame(data)
df3
```

Out[252…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [253…

```
df3.bfill()
```

Out[253…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	99.0	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	84.0	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [254…

```
df4 = pd.DataFrame(data)
df4
```

Out[254…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [255…

```
df4["Subject 3"] = df4["Subject 3"].fillna(75)
df4["Subject 4"] = df4["Subject 4"].fillna(85)
df4
```

Out[255…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	85.0	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	75.0	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [256…

```
df5 = pd.DataFrame(data)
df5
```

Out[256…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [257…

```
df5["Subject 3"] = df5["Subject 3"].fillna(df5["Subject 3"].mean())
df5["Subject 4"] = df5["Subject 4"].fillna(df5["Subject 4"].mean())
df5
```

Out[257…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.00	92.000	96
1	2	Student_2	72	97	82.00	80.545	78
2	3	Student_3	100	88	71.00	99.000	-94
3	4	Student_4	72	99	79.89	84.000	86
4	5	Student_5	97	70	84.00	70.000	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.00	66.000	98
997	998	Student_998	61	96	62.00	84.000	83
998	999	Student_999	72	76	90.00	72.000	90
999	1000	Student_1000	68	87	100.00	76.000	79
1000	1	Student_1	100	62	73.00	92.000	96

1001 rows × 7 columns

In [258…

```
df6 = pd.DataFrame(data)
df6
```

Out[258…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [259…

```
df6["Subject 3"] = df6["Subject 3"].fillna(df6["Subject 3"].median())
df6["Subject 4"] = df6["Subject 4"].fillna(df6["Subject 4"].median())
df6
```

Out[259…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	81.0	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	80.0	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [260…

```
df7 = pd.DataFrame(data)
df7
```

Out[260…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [261…

```
nc = ['Subject 1', 'Subject 2', 'Subject 3', 'Subject 4', 'Attendance']
```

In [262…

```
for c in nc:
    df7[c] = df7[c].fillna(df7[c].mean())
df7
```

Out[262…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.00	92.000	96
1	2	Student_2	72	97	82.00	80.545	78
2	3	Student_3	100	88	71.00	99.000	-94
3	4	Student_4	72	99	79.89	84.000	86
4	5	Student_5	97	70	84.00	70.000	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.00	66.000	98
997	998	Student_998	61	96	62.00	84.000	83
998	999	Student_999	72	76	90.00	72.000	90
999	1000	Student_1000	68	87	100.00	76.000	79
1000	1	Student_1	100	62	73.00	92.000	96

1001 rows × 7 columns

In [263…

```
df8=pd.DataFrame(data)
df8
```

Out[263…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [264…

```
nc = ['Subject 1', 'Subject 2', 'Subject 3', 'Subject 4', 'Attendance']
```

In [265…

```
for c in nc:
    df8[c] = df8[c].fillna(df8[c].mean())
df8
```

Out[265…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.00	92.000	96
1	2	Student_2	72	97	82.00	80.545	78
2	3	Student_3	100	88	71.00	99.000	-94
3	4	Student_4	72	99	79.89	84.000	86
4	5	Student_5	97	70	84.00	70.000	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.00	66.000	98
997	998	Student_998	61	96	62.00	84.000	83
998	999	Student_999	72	76	90.00	72.000	90
999	1000	Student_1000	68	87	100.00	76.000	79
1000	1	Student_1	100	62	73.00	92.000	96

1001 rows × 7 columns

In [266…

```
df9=pd.DataFrame(data)
df9
```

Out[266…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [267…

```
nc = ['Subject 1', 'Subject 2', 'Subject 3', 'Subject 4', 'Attendance']
```

In [268…

```
for c in nc:
    df9[c] = df9[c].fillna(df9[c].median())
df9
```

Out[268…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	81.0	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	80.0	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [269…

```
df10 = pd.DataFrame(data)
df10
```

Out[269…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [270…

```
df10.replace(to_replace=np.nan, value=85)
```

Out[270…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	85.0	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	85.0	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [271…

```
df11=pd.DataFrame(data)
df11
```

Out[271…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [272…

```
df11[df11.select_dtypes(include='number').columns] = (
    df11.select_dtypes(include='number')
        .interpolate(method='linear', limit_direction='forward')
)
df11
```

Out[272…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	95.5	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	77.5	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [273…

```
df12 = pd.DataFrame(data)
df12
```

Out[273…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [274…

```
df12.dropna()
```

Out[274…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
2	3	Student_3	100	88	71.0	99.0	-94
4	5	Student_5	97	70	84.0	70.0	86
5	6	Student_6	98	76	89.0	92.0	82
6	7	Student_7	61	64	97.0	98.0	83
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

999 rows × 7 columns

In [275…

```
df13=pd.DataFrame(data)
df13
```

Out[275…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [276…

```
df13.dropna(how='all', inplace=True)
df13
```

Out[276…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [277…

```
df13=pd.DataFrame(data)
df13
```

Out[277…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [278…

```
df13.dropna(axis=1)
```

Out[278…

	Roll No	Name	Subject 1	Subject 2	Attendance
0	1	Student_1	100	62	96
1	2	Student_2	72	97	78
2	3	Student_3	100	88	-94
3	4	Student_4	72	99	86
4	5	Student_5	97	70	86
...	...	...	...	...	...
996	997	Student_997	88	68	98
997	998	Student_998	61	96	83
998	999	Student_999	72	76	90
999	1000	Student_1000	68	87	79
1000	1	Student_1	100	62	96

1001 rows × 5 columns

In [279…

```
df14=pd.DataFrame(data)
df14
```

Out[279…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [280…

```
df14.dropna(axis=0, how='any')
```

Out[280…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
2	3	Student_3	100	88	71.0	99.0	-94
4	5	Student_5	97	70	84.0	70.0	86
5	6	Student_6	98	76	89.0	92.0	82
6	7	Student_7	61	64	97.0	98.0	83
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

999 rows × 7 columns

In [281…

```
df15=pd.DataFrame(data)
df15
```

Out[281…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [282…

```
df15[df15[['Subject 1', 'Subject 2', 'Subject 3','Attendance']]<0]
```

Out[282…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN	NaN	NaN	-94.0
3	NaN	NaN	NaN	NaN	NaN	NaN	NaN
4	NaN	NaN	NaN	NaN	NaN	NaN	NaN
...	...	...	...	...	...	...	...
996	NaN	NaN	NaN	NaN	NaN	NaN	NaN
997	NaN	NaN	NaN	NaN	NaN	NaN	NaN
998	NaN	NaN	NaN	NaN	NaN	NaN	NaN
999	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1000	NaN	NaN	NaN	NaN	NaN	NaN	NaN

1001 rows × 7 columns

In [283…

```
df16 = pd.DataFrame(data)
df16
```

Out[283…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [284…

```
df16['Attendance']=df16['Attendance'].clip(lower=0)
df16
```

Out[284…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	0
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [285…

```
df17=pd.DataFrame(data)
df17
```

Out[285…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [286…

```
df17.drop_duplicates(inplace=True)
df17
```

Out[286…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
995	996	Student_996	74	89	85.0	71.0	87
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79

1000 rows × 7 columns

In [287…

```
import seaborn as sns
import matplotlib.pyplot as plt
```

In [288…

```
df18=pd.DataFrame(data)
df18
```

Out[288…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.0	92.0	96
1	2	Student_2	72	97	82.0	NaN	78
2	3	Student_3	100	88	71.0	99.0	-94
3	4	Student_4	72	99	NaN	84.0	86
4	5	Student_5	97	70	84.0	70.0	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.0	66.0	98
997	998	Student_998	61	96	62.0	84.0	83
998	999	Student_999	72	76	90.0	72.0	90
999	1000	Student_1000	68	87	100.0	76.0	79
1000	1	Student_1	100	62	73.0	92.0	96

1001 rows × 7 columns

In [289…

```
for c in nc:
    df18[c] = df18[c].fillna(df18[c].mean())
df18
```

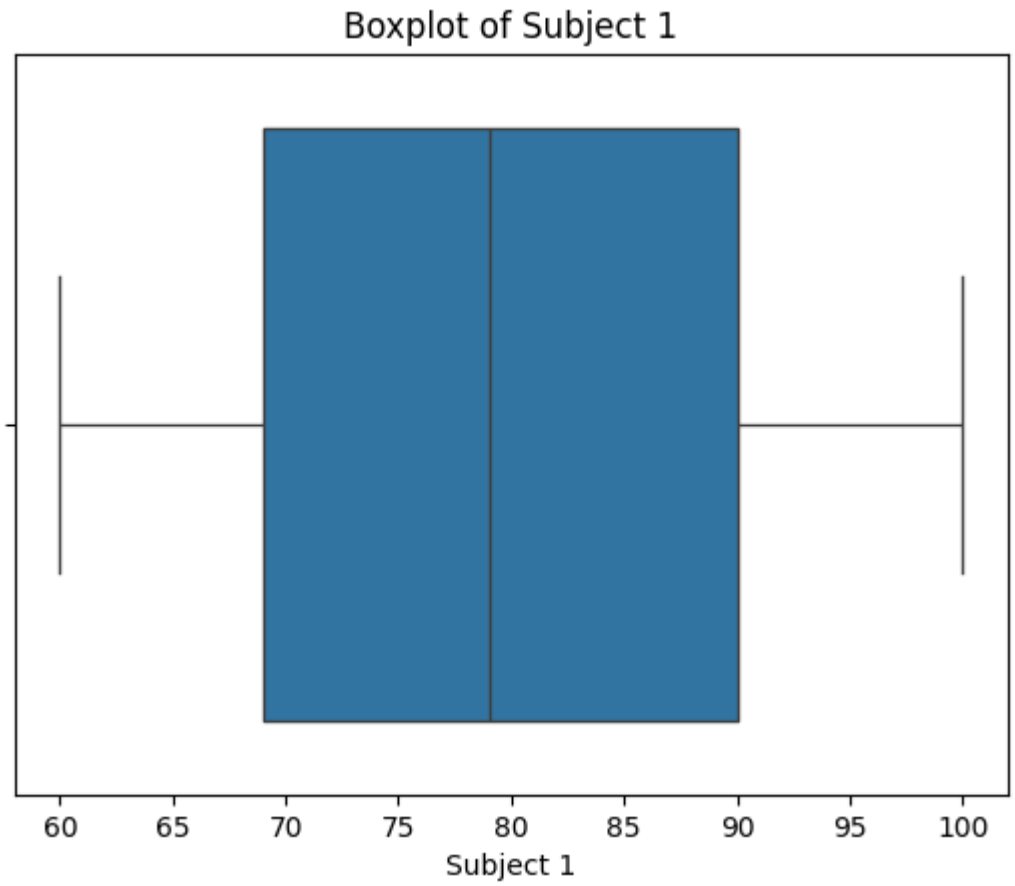
Out[289…

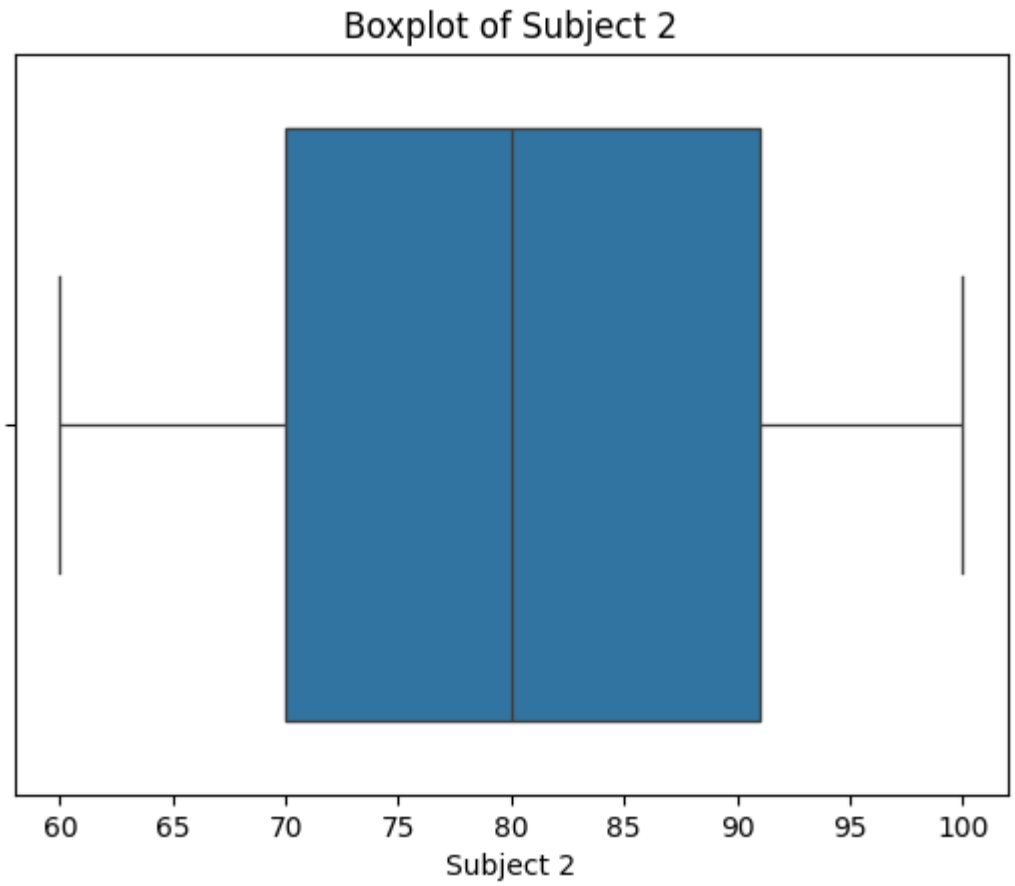
	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100	62	73.00	92.000	96
1	2	Student_2	72	97	82.00	80.545	78
2	3	Student_3	100	88	71.00	99.000	-94
3	4	Student_4	72	99	79.89	84.000	86
4	5	Student_5	97	70	84.00	70.000	86
...	...	...	...	...	...	...	...
996	997	Student_997	88	68	84.00	66.000	98
997	998	Student_998	61	96	62.00	84.000	83
998	999	Student_999	72	76	90.00	72.000	90
999	1000	Student_1000	68	87	100.00	76.000	79
1000	1	Student_1	100	62	73.00	92.000	96

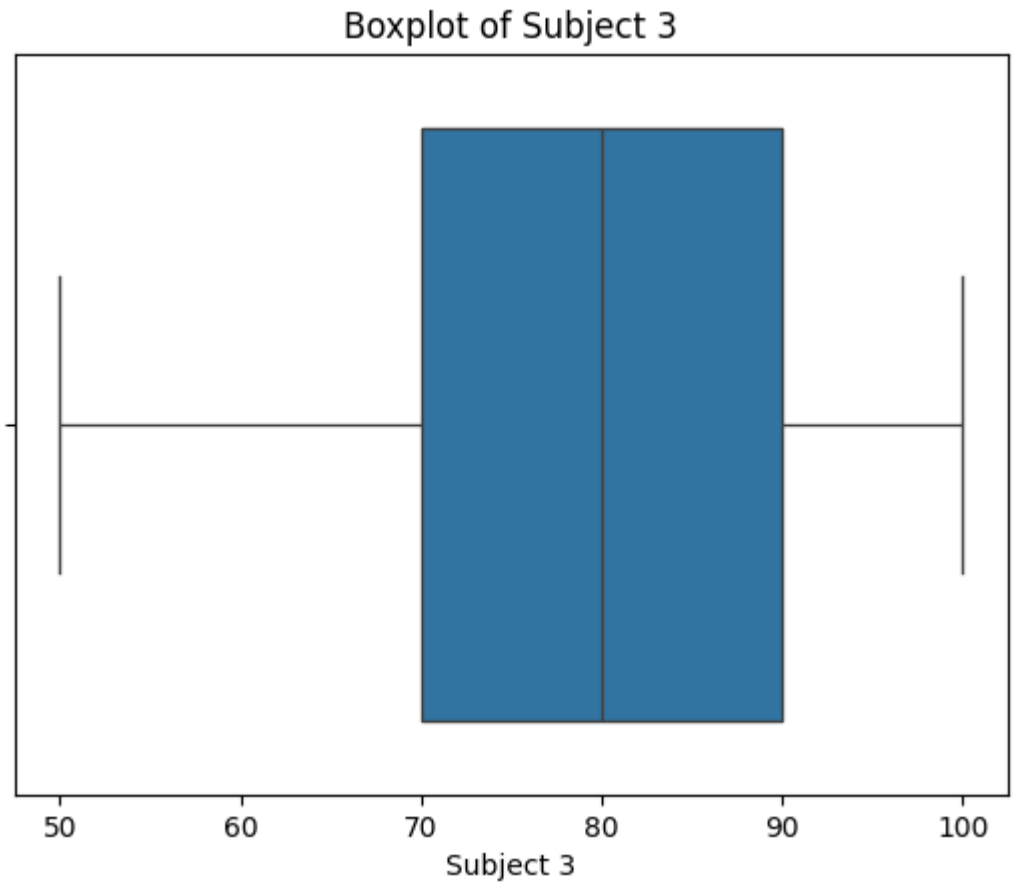
1001 rows × 7 columns

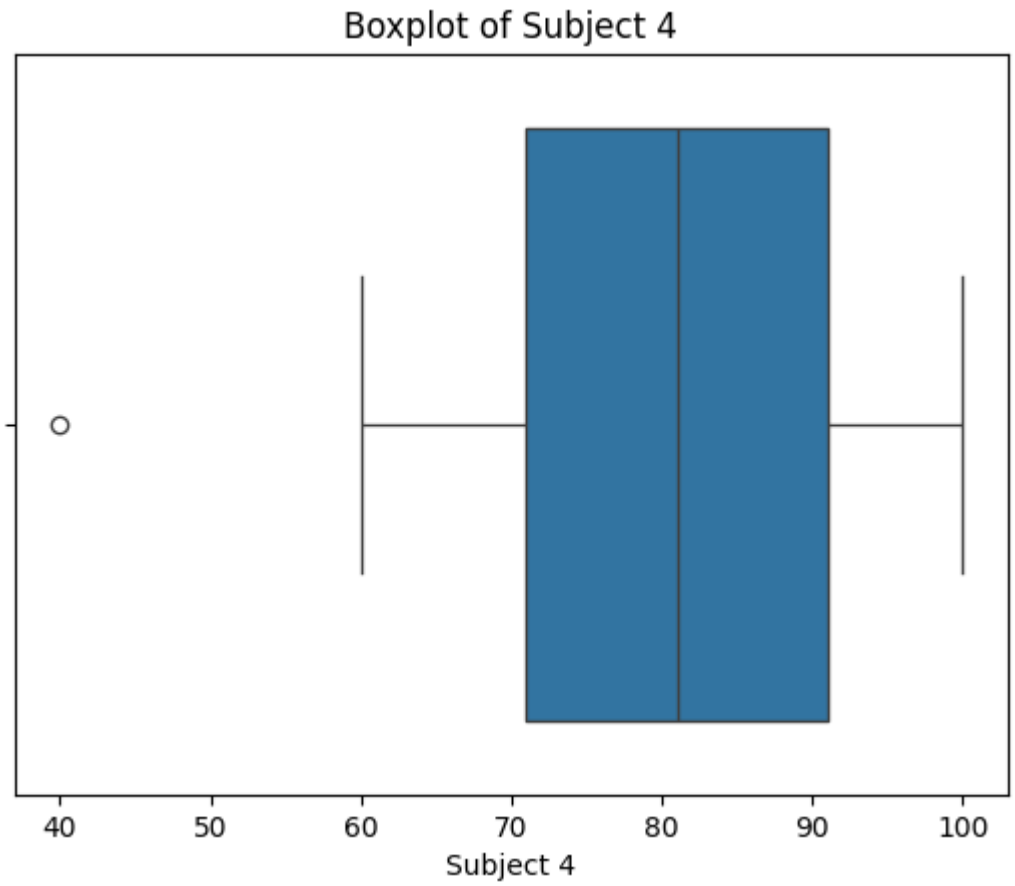
In [290…

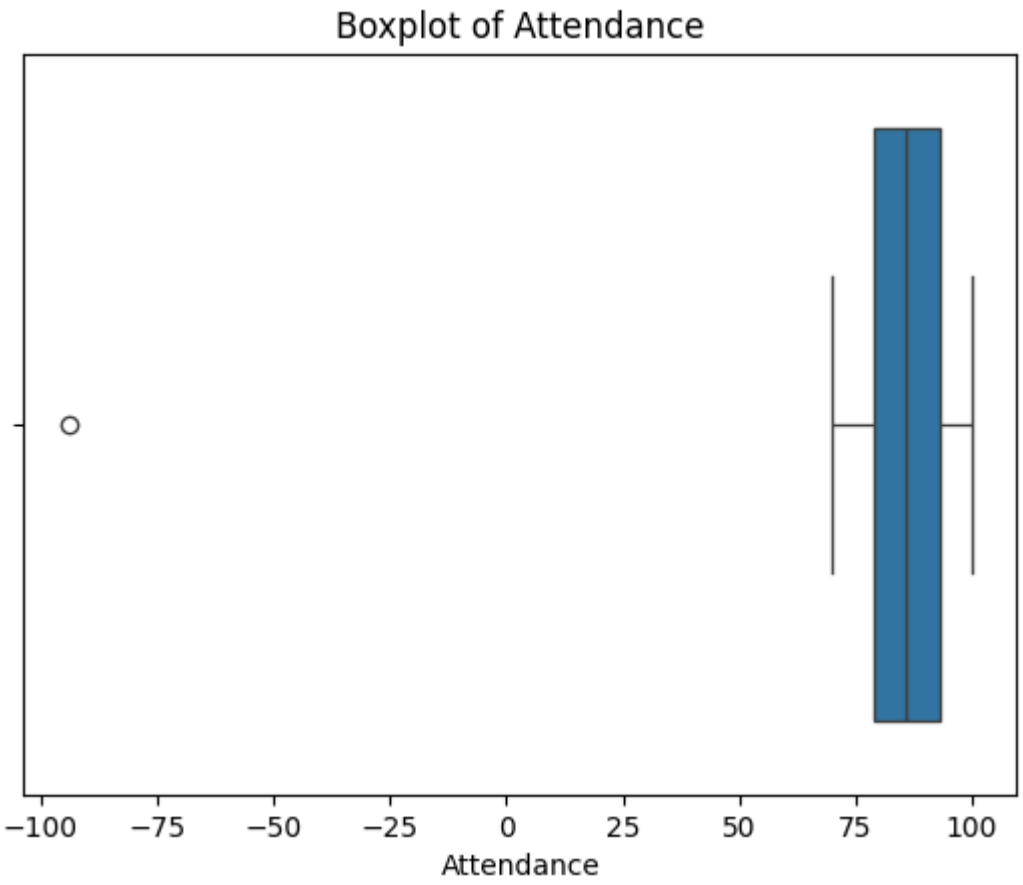
```
for c in nc:
    sns.boxplot(x = df18[c])
    plt.title(f"Boxplot of {c}")
    plt.show()
```


Boxplot of Subject 1
60
65
70
75
80
85
90
95
100
Subject 1


Boxplot of Subject 2
60
65
70
75
80
85
90
95
100
Subject 2


Boxplot of Subject 3
50
60
70
80
90
100
Subject 3


Boxplot of Subject 4
40
50
60
70
80
90
100
Subject 4



In [291…

```
for c in nc:
    Q1 = df18[c].quantile(0.25)
    Q3 = df18[c].quantile(0.75)
    IQR = Q3 - Q1
    lb = Q1 - 1.5 * IQR
    ub = Q3 + 1.5 * IQR
    df18[c] = np.where((df18[c] < lb) | (df18[c] > ub), df18[c].median(), df18[c])
df18
```

Out[291…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance
0	1	Student_1	100.0	62.0	73.00	92.000	96.0
1	2	Student_2	72.0	97.0	82.00	80.545	78.0
2	3	Student_3	100.0	88.0	71.00	99.000	86.0
3	4	Student_4	72.0	99.0	79.89	84.000	86.0
4	5	Student_5	97.0	70.0	84.00	70.000	86.0
...	...	...	...	...	...	...	...
996	997	Student_997	88.0	68.0	84.00	66.000	98.0
997	998	Student_998	61.0	96.0	62.00	84.000	83.0
998	999	Student_999	72.0	76.0	90.00	72.000	90.0
999	1000	Student_1000	68.0	87.0	100.00	76.000	79.0
1000	1	Student_1	100.0	62.0	73.00	92.000	96.0

1001 rows × 7 columns

In [292…

```
from scipy import stats
z = np.abs(stats.zscore(df18["Subject 4"]))
print(z)
```

```
[0.97456486 0.00349721 1.57224574 ... 0.73309479 0.39156286 0.97456486]
```

In [293…

```
df18["Attendance"] = pd.to_numeric(df18["Attendance"], errors="coerce")

df18["log_attendance"] = np.log1p(
    df18["Attendance"].clip(lower=0)
)
df18
```

Out[293…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance	log_attendance
0	1	Student_1	100.0	62.0	73.00	92.000	96.0	4.574711
1	2	Student_2	72.0	97.0	82.00	80.545	78.0	4.369448
2	3	Student_3	100.0	88.0	71.00	99.000	86.0	4.465908
3	4	Student_4	72.0	99.0	79.89	84.000	86.0	4.465908
4	5	Student_5	97.0	70.0	84.00	70.000	86.0	4.465908
...	...	...	...	...	...	...	...	...
996	997	Student_997	88.0	68.0	84.00	66.000	98.0	4.595120
997	998	Student_998	61.0	96.0	62.00	84.000	83.0	4.430817
998	999	Student_999	72.0	76.0	90.00	72.000	90.0	4.510860
999	1000	Student_1000	68.0	87.0	100.00	76.000	79.0	4.382027
1000	1	Student_1	100.0	62.0	73.00	92.000	96.0	4.574711

1001 rows × 8 columns

In [294…

```
df18["Subject 2"] = df18["Subject 2"].astype(int)
df18
```

Out[294…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance	log_attendance
0	1	Student_1	100.0	62	73.00	92.000	96.0	4.574711
1	2	Student_2	72.0	97	82.00	80.545	78.0	4.369448
2	3	Student_3	100.0	88	71.00	99.000	86.0	4.465908
3	4	Student_4	72.0	99	79.89	84.000	86.0	4.465908
4	5	Student_5	97.0	70	84.00	70.000	86.0	4.465908
...	...	...	...	...	...	...	...	...
996	997	Student_997	88.0	68	84.00	66.000	98.0	4.595120
997	998	Student_998	61.0	96	62.00	84.000	83.0	4.430817
998	999	Student_999	72.0	76	90.00	72.000	90.0	4.510860
999	1000	Student_1000	68.0	87	100.00	76.000	79.0	4.382027
1000	1	Student_1	100.0	62	73.00	92.000	96.0	4.574711

1001 rows × 8 columns

In [295…

```
col_min = df18["Subject 1"].min()
col_max = df18["Subject 1"].max()
df18["Scaled_Data_Manual"] = (df18["Subject 1"] - col_min) / (col_max - col_min)
```

In [296…

```
df18
```

Out[296…

	Roll No	Name	Subject 1	Subject 2	Subject 3	Subject 4	Attendance	log_attendance	Scaled_Data_Manual
0	1	Student_1	100.0	62	73.00	92.000	96.0	4.574711	1.000
1	2	Student_2	72.0	97	82.00	80.545	78.0	4.369448	0.300
2	3	Student_3	100.0	88	71.00	99.000	86.0	4.465908	1.000
3	4	Student_4	72.0	99	79.89	84.000	86.0	4.465908	0.300
4	5	Student_5	97.0	70	84.00	70.000	86.0	4.465908	0.925
...	...	...	...	...	...	...	...	...	...
996	997	Student_997	88.0	68	84.00	66.000	98.0	4.595120	0.700
997	998	Student_998	61.0	96	62.00	84.000	83.0	4.430817	0.025
998	999	Student_999	72.0	76	90.00	72.000	90.0	4.510860	0.300
999	1000	Student_1000	68.0	87	100.00	76.000	79.0	4.382027	0.200
1000	1	Student_1	100.0	62	73.00	92.000	96.0	4.574711	1.000

1001 rows × 9 columns

In [ ]:

In [ ]: